

Unit 7, Lesson 6: Special Right Triangles



Benchmark

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



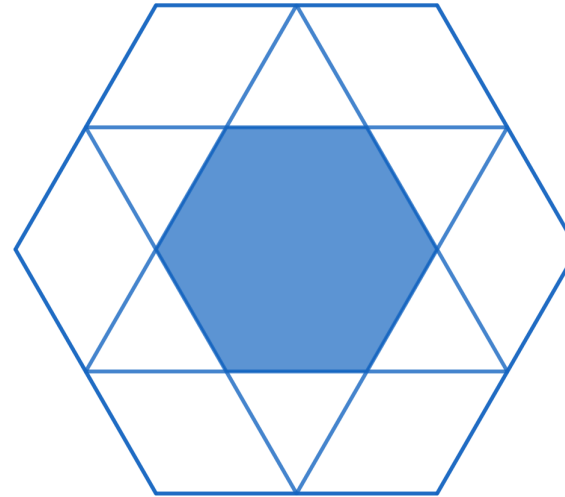
Learning Targets

- ☐ I can use side ratios of 45 – 45 – 90 triangles to solve problems.
- ☐ I can use side ratios of 30 – 60 – 90 triangles to solve problems.



7.6.1 Warm-Up: Critical Thinking

1. What fraction of the shape is shaded?

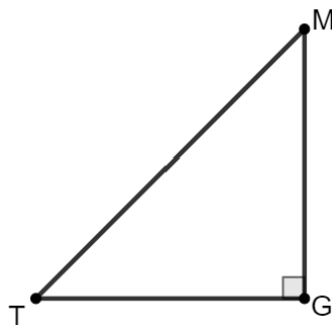


2. Describe your problem-solving method.



7.6.2 Exploration: Exploring Special Right Triangles

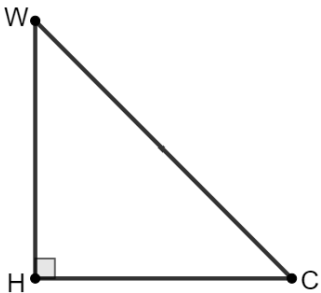
1. $\triangle MGT$ is a right triangle, where $m\angle MGT = 90^\circ$, $MG = 6$, and $GT = 6$.



- a. What type of right triangle is $\triangle MGT$?
- b. Determine the measurements. Leave the measurement of $\overline{MT}$ in radical form.

MT	$m\angle GMT$	$m\angle GTM$

2. $\triangle WCH$ is a right triangle, where $m\angle WHC = 90^\circ$, $m\angle WCH = 45^\circ$, and $WC = 4\sqrt{2}$.



- a. What type of right triangle is $\triangle WCH$?
- b. Determine the measurements.

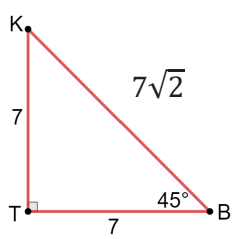
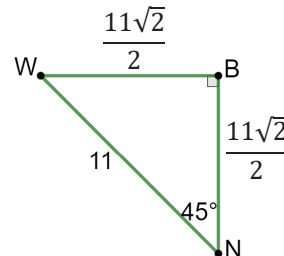
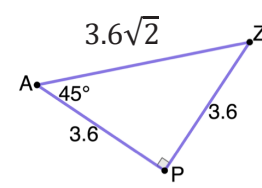
HC	WH	$m\angle CWH$

3. What do you notice about the angles of $\triangle WCH$ and $\triangle MGT$?

4. What do you notice about the legs of $\triangle WCH$ and $\triangle MGT$?

$\triangle WHC$ and $\triangle MGT$ are $45 - 45 - 90$ triangles.

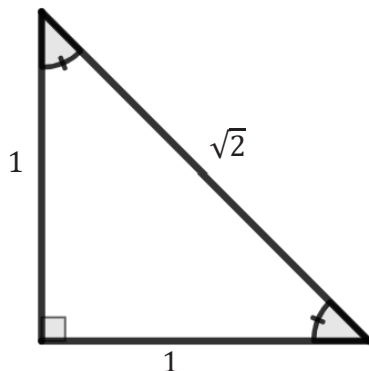
5. The table shows three $45 - 45 - 90$ triangles.

$\triangle KBT$	$\triangle WBN$	$\triangle AZP$
		

- a. Discuss with your partner the relationship between the legs of a $45 - 45 - 90$ triangle and its hypotenuse. Write a conjecture.
- b. Ask two other pairs of partners for the conjecture they wrote. Listen to their conjecture and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

Conjecture	Initials

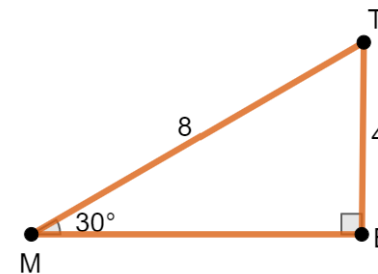
6. An isosceles right triangle is shown.



Describe how the triangle shows the relationships in a 45 – 45 – 90 triangle.

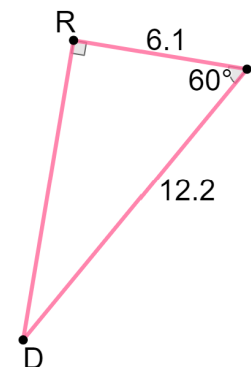
7. $\triangle MBT$ is a right triangle, where $m\angle MBT = 90^\circ$, $m\angle TMB = 30^\circ$, $TB = 4$, and $MT = 8$.

Find the length of $\overline{MB}$ using the Pythagorean Theorem. Write the answer in simplified radical form.



8. $\triangle RFD$ is a right triangle, where $m\angle DRF = 90^\circ$, $m\angle RFD = 60^\circ$, $RF = 6.1$, and $FD = 12.2$.

Discuss with your partner what features $\triangle RFD$ has in common with $\triangle MBT$.



9. $\triangle MBT$ and $\triangle RFD$ are 30 – 60 – 90 triangles.

The table gives the exact measurements for each of the sides of $\triangle MBT$ and $\triangle RFD$.

Triangle	Short Leg	Long Leg	Hypotenuse
$\triangle MBT$	4	$4\sqrt{3}$	8
$\triangle RFD$	6.1	$6.1\sqrt{3}$	12.2

Complete the table by describing the relationship between the side lengths of a 30 – 60 – 90 triangle.

Sides	Relationship
short leg and hypotenuse	
short leg and long leg	

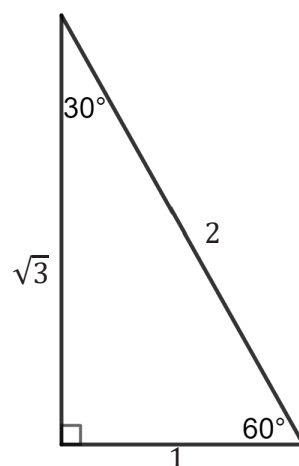
10. The table below includes three 30 – 60 – 90 triangles.

$\triangle KBL$	
$\triangle VHK$	
$\triangle HRY$	

Discuss with your partner the relationships between the sides of a 30 – 60 – 90 triangle in the table. Complete the table by showing how the short side of each of the triangles can be used to determine the long side and the hypotenuse. Write the final length as an exact value.

Triangle	Short Side	Work to Find the Long Side	Work to Find the Hypotenuse
ΔKBL	$4\sqrt{3}$		
ΔVHK	$\frac{7}{\sqrt{3}}$		
ΔHRY	$2\sqrt{7}$		

11. Describe how the triangle shows the relationships in a 30 – 60 – 90 triangle.

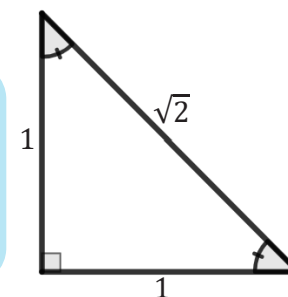


7.6.3 Guided Instruction: 45-45-90 and 30-60-90 Triangles



The side lengths of a 45 – 45 – 90 triangle can be found using the equation

$$\text{leg} \cdot \sqrt{2} = \text{hypotenuse}.$$



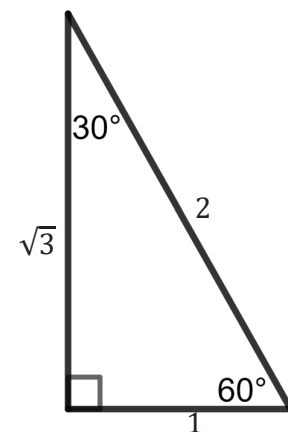
1. ΔZYN has $m\angle YZN = m\angle YNZ = 45^\circ$, $m\angle ZYN = 90^\circ$, and $ZN = 12$. What are the lengths of the legs of the triangle? Leave your answer as an exact answer.



The lengths of the short leg and the long leg of a 30 – 60 – 90 triangle can be found using the equation $\text{short leg} \cdot \sqrt{3} = \text{long leg}$.



The lengths of the short leg and the hypotenuse of a 30 – 60 – 90 triangle can be found using the equation $\text{short leg} \cdot 2 = \text{hypotenuse}$.



2. $\triangle BHK$ is a $30 - 60 - 90$ triangle with $m\angle KBH = 30^\circ$, $m\angle BKH = 90^\circ$, and $BK = 5\sqrt{3}$. What are the measures of the other sides of the triangle?



7.6.4 Guided Practice: Solving Special Right Triangles

1. $\triangle LPM$ is a $45 - 45 - 90$ triangle with $m\angle LMP = m\angle MLP = 45^\circ$, $m\angle LPM = 90^\circ$, and $LM = 7$. What are the lengths of $\overline{PM}$ and $\overline{LP}$?
2. $\triangle TNW$ is a right triangle with $m\angle NTW = 60^\circ$, $m\angle WNT = 90^\circ$, and $TW = 18$. What are the measures of the other sides of the triangle?



7.6.5 Wrap-Up: Ticket Out the Door

1. $\triangle FHX$ has angle measures $m\angle HXF = 90^\circ$ and $m\angle HFX = 30^\circ$, and side length $HX = 6\sqrt{3}$. What are the exact measures of $\overline{FX}$ and $\overline{FH}$?

Unit 7, Lesson 7: The Unit Circle and Its Coordinates



Benchmarks

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



Learning Targets

- ☐ I can explore relationships between the coordinates of vertices when the triangles are in different quadrants.
- ☐ I can use side ratios of 45 – 45 – 90 triangles to solve problems.
- ☐ I can use side ratios of 30 – 60 – 90 triangles to solve problems.